

Lab 1: Classical Ciphers

Requirement 1: Playfair Cypher [Hard]

1.1- Implement Playfair cypher function:

- Accepts parameters:-
 - **Key** (String; representing the key)
 - **Enc_dec_flag** (Int; 1 for encryption or 2 for decryption)
 - **Text** (String; be it either the ciphertext in case of decryption or the plaintext in case of encryption)
- Prints the Playfair matrix
- Returns a String corresponding to the deciphered plaintext or encrypted ciphertext.

1.2- Implement a Brute Force algorithm to get *meaningful* sentences:

- Tries all possible permutations of keys of length (n) ~n will be small
- You will be supplied with a text file containing the top 1000 most used English words.
- Search for any matches between the dictionary and the candidate plaintext.
- Print all plaintexts containing at least 1 match.

M	O	N	A	R
C	H	Y	B	D
E	F	G	I/J	K
L	P	Q	S	T
U	V	W	X	Z

Assumptions:

- 1- Spaces are **not encrypted**
- 2- Word boundaries are spaces (this will make the brute force search easier)
- 3- Key length for brute force ≤ 3 (but for enc/dec, the key length ≤ 10)
- 4- All text is lowercase
- 5- Convert all "j"s to "i"s
- 6- If the length of the plaintext after padding with "x"s is odd, add a dummy "z" at the end of it (ex: speedy => sp ex ed yz)